Survey of the Human Proteostasis Network: Translation, Protein Folding, and Organelle-Specific Systems

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Abstract

The condition of having a healthy, functional proteome is known as protein homeostasis, or proteostasis. Maintaining proteostasis is the province of the proteostasis network, which consists of the cellular components that regulate protein synthesis, folding, localization, and degradation. The proteostasis network is a fundamental functional unit in biology with relevance to many diseases. Despite its importance, the human proteostasis network has until now lacked a unified definition, limiting our ability to interrogate proteostasis systematically, integrate multiomic datasets, compare results across studies, or model proteostasis dysfunction in disease. To address this limitation, we have operationally defined the human proteostasis network by providing a comprehensive, annotated and manually curated list of its components classified into nine Branches based on their function and localization: cytonuclear proteostasis, ER proteostasis, mitochondrial proteostasis, nuclear proteostasis, proteostasis network regulation, translation, extracellular proteostasis, the autophagy lysosome pathway, and the ubiquitin proteasome system; which we present in a series of three manuscripts of which this is the first. Here, we present a curated list of over 1100 components that make up the first seven of these Branches, which comprise the compartment-based proteostasis network (cytonuclear, ER, mitochondrial, nuclear, and extracellular proteostasis), proteostasis network regulation, and translation. These Branches include the cell's translation machinery, chaperones, folding enzymes, systems for trafficking proteins, and organelle-specific degradation systems. This work provides a framework for interrogating proteostasis at scale, which will facilitate mechanistic insights into how proteostasis is preserved in health, lost in aging and disease, and how it could be restored through pharmacological intervention.

Keywords

Annotation; chaperones; cellular stress responses; protein synthesis; protein folding; protein transport

Abbreviations

ALP: Autophagy–lysosome pathway; ATF4: Activating transcription factor 4; ATF6: Activating transcription factor 6; BAG: BCL2-associated athanogene; BBSome: Bardet–Biedl syndrome protein complex; CC-TPR: Carboxylate clamp tetratricopeptide repeat; CCT / TRiC: Chaperonin containing TCP1 / T-complex protein ring complex; CLU: Clusterin; COPI: Coat protein complex I; COPII: Coat protein complex II; ER: Endoplasmic reticulum; ERAD: Endoplasmic reticulum–associated degradation; EMC: ER membrane protein complex; GET: Guided entry of tail-anchored proteins; GRP170: Glucose-regulated protein 170; GO: Gene Ontology; HSF1: Heat shock factor 1; HSP: Heat shock protein; HSP70: 70-kDa heat shock protein; HSP90: 90-kDa heat shock protein; HSPBP1: HSPA binding protein 1; HSP110: 110-kDa heat shock protein; ISR: Integrated stress response; KEAP1: Kelch-like ECH-associated protein 1; KDEL: Lys-Asp-Glu-Leu ER retention signal; mRNA: Messenger RNA; NAC: Nascent polypeptide-associated complex; NEF: Nucleotide exchange factor; NFE2L2 / NRF2: Nuclear factor erythroid 2-related factor 2;NPC: Nuclear pore complex; OST: Oligosaccharyl transferase; PDI: Protein disulfide isomerase; PERK: Protein kinase RNA-like ER kinase (EIF2AK3); PN: Proteostasis network; PPIase: Peptidyl-prolyl cis–trans isomerase; RAN: Ras-related nuclear protein; RQC: Ribosome-associated quality control; SRP: Signal recognition particle; TPR: Tetratricopeptide repeat; tRNA: Transfer RNA; TOMM: Translocase of the outer mitochondrial membrane; TIMM: Translocase of the inner mitochondrial membrane; UPR: Unfolded protein response; UPS: Ubiquitin–proteasome system; VCP: Valosin-containing protein; XBP1: X-box binding protein 1

Introduction

The process by which nascent proteins fold into their native structures co- or post-translationally is imperfect and can fail [1-7]. These failures leave proteins in non-native states that are nonfunctional and potentially toxic, especially if they form aggregates that alter the stoichiometry of macromolecular complexes or interfere with fundamental cellular processes [3,6,8-10]. To maintain the proteome in a healthy, functional state—a condition known as protein homeostasis, or proteostasis—all organisms, beginning at least with the last universal common ancestor [3,11,12], have cellular components that manage the synthesis, folding, trafficking, and degradation of proteins. These components are collectively designated as the proteostasis network [3-5,13,14].

The proteostasis network, while widely discussed in the literature, remains loosely defined because there has been no systematic and comprehensive account of its components. The efforts made toward this end have mostly focused on chaperones [15,16], with less effort devoted to other equally important arms of the proteostasis network involved in protein transport and degradation. The widespread interest in diseases characterized by failures of proteostasis—especially aging-related neurodegenerative diseases [17-21] but in many other diseases as well [22]—and intensive efforts to identify critical limiting targets for therapeutic interventions suggest that having such a comprehensive description of this network is imperative. We address this gap in our knowledge in this and subsequent papers in this series by presenting a comprehensive and structured framework of the human proteostasis network.

In this paper, we focus on chaperones, folding enzymes, systems for trafficking proteins into and out of organelles, and the machinery of protein synthesis. To create a comprehensive and reliable list, we combined exhaustive manual literature curation with homology-based analyses, allowing us to include not only well-characterized components but also proteins whose sequence or structural features strongly predict a role in proteostasis. This systematic approach ensures that the resulting network captures both canonical and less obvious contributors, providing a robust foundation for future studies. The next two manuscripts in this series will present the annotation of the two large and complex systems for protein degradation in eukaryotes, namely the autophagy-lysosome pathway (ALP) and the ubiquitin-proteasome system (UPS). This foundational

classification provides the basis for our subsequent systematic definition of proteostasis networks in genetically tractable experimental systems.

Results

Overview of the proteostasis network

The proteostasis network consists of the systems that regulate and manage protein synthesis, folding, localization, and degradation [3-5,13,14]. The components responsible for cytosolic and mitochondrial protein synthesis (Figure 1) include the ribosome, ribosome biogenesis factors, and translation initiation, elongation, and termination factors. Both the cytosol and the mitochondria have their own complements of these components (with some overlap among tRNA synthetases). The cytosol additionally contains components responsible for ribosomal quality control (RQC). The components responsible for protein folding include the canonical chaperone systems (HSP70s, HSP90s, small HSPs, and TRiC/CCT, and their associated co-chaperones); the folding enzymes (peptidyl-prolyl isomerases and protein disulfide isomerases), some of which also function as co-chaperones; the glycoprotein folding machinery of the endoplasmic reticulum (ER), ranging from oligosaccharyl transferase to the lectin chaperones to the folding-state-responsive glycan trimming enzymes; and other, more substrate- or organelle-specific folding systems (e.g., for collagen). The components responsible for protein localization to organelles like the ER, the mitochondria, and the nucleus include the channels through which proteins are shuttled into and out of organelles and the components that recognize substrates for transport. Finally, the components of the degradation pathways include the organelle-specific degradation systems in the ER (ER-associated degradation, or ERAD) and mitochondria (mitochondrial proteases), the ALP and the UPS. However, the ALP and the UPS, which serve more broadly in the cell, will be addressed in subsequent papers in this series. We have excluded processes that are essential for protein biogenesis and transport but with only indirect influence on proteostasis such as amino acid synthesis and core metabolism (which produce amino acids for protein synthesis), gene transcription and mRNA quality control (which influence proteome composition), and cytoskeletal transport (which contributes to protein localization).

Criteria for inclusion in the proteostasis network

We used "entity-based" and "domain-based" criteria to determine whether a given component should be included in the proteostasis network. "Entity-based" refers to inclusion of an "entity" (a protein or non-coding RNA) based on positive biochemical, cell biological, or genetic evidence in the literature that it functions in proteostasis. "Domain-based" refers to inclusion based on a protein containing a proposed structural domain that is strongly associated with proteostasis. Preliminary lists of proteostasis network components were generated based on reviews and other literature on the relevant systems (see Methods). These preliminary lists were then manually vetted gene by gene by the authors of the current study. For borderline cases, rationales for inclusion or exclusion were presented by Proteostasis Consortium members with subject area expertise and a decision was made based on the consensus of the group.

An instructive example of entity-based inclusion is the protein VCP (valosin-containing protein; also known as p97). VCP is a member of the AAA protein family (ATPases Associated with diverse cellular Activities). This large family of proteins contains some members that are involved in proteostasis but many that are not; thus, being a AAA protein does not imply being a proteostasis network component. However, VCP has many demonstrated roles in proteostasis; in fact, it has more roles than any other component of the proteostasis network. VCP is known to be necessary for ERAD [23], mitochondrial translocation-associated degradation (mitoTAD) [24], and ribosome-associated quality control [25], not to mention its roles in the ALP [26] and the UPS [27]. These experimentally documented functions in proteostasis qualify VCP to be included as a component of the proteostasis network.

A representative example of domain-based inclusion is provided by the HSP70 chaperones. The HSP70s are central to proteostasis in almost all organisms (excepting the thermophilic archaea) [11] and have a conserved, characteristic domain structure, consisting of an N-terminal nucleotide binding domain and a C-terminal substrate binding domain [28,29]. Humans have 12 HSP70 genes (excluding those that encode HSPs fused to other large domains, like the HSP110s and HYOU1) [30]. Many of them have central, and abundantly documented, roles in proteostasis, like HSPA8 (the constitutively expressed cytonuclear HSP70), HSPA1A/HSPA1B (the stress-responsive cytonuclear HSP70s), HSPA5 (the ER-resident HSP70), and HSPA9 (the mitochondrial HSP70) [28]; however, this is not true for all human HSP70s. HSPA12A and

HSPA12B in particular are very poorly annotated and their functions in proteostasis are not yet known as of this writing. However, the HSP70 domain architecture is so closely associated with proteostasis that we are essentially assuming that it is more likely than not that such functions will eventually be uncovered; however, should future evidence contradict this, these components will be removed from the network.

Finally, we end with an example of how the entity- and domain-based systems have sometimes been used in conjunction. TPR (tetratricopeptide repeat) domain-containing proteins can bind to both HSP70 and HSP90 cytosolic chaperones and therefore serve as cochaperones that integrate the activity of both chaperone systems. However, only a subset of TPR domain-containing proteins, those that contain a "carboxylate clamp", can serve in this role [31]; these are denoted CC-TPR domains. The carboxylate clamp is a structural motif with two basic residues that bind to the carboxylate residues in the EEVD sequences at the C-termini of cytosolic HSP70 and HSP90 chaperones [32]. Of the >100 cytosolic TPR domain-containing proteins [15], only 31 have the carboxylate clamp motif and are considered true HSP70/HSP90 cochaperones and therefore belong in the proteostasis network [31]. Thus, we had to refine the domain-based approach using information from the literature to provide an accurate accounting of the proteostasis network in this case.

Taxonomic scheme for annotation

To organize the proteostasis network components, we sought an annotation system that would convey at a glance a component's localization and function in proteostasis. Such an approach complements other formalized annotation systems that rely on structured vocabularies, such as the Gene Ontology [33,34]. We developed a simple taxonomic scheme consisting of five levels: Branch, Class, Group, Type, and Subtype. We found that five levels were sufficient to convey a general sense of each component's localization and function while minimizing the number of descriptors. The broadest category, Branch, refers to a component's localization or membership in an overarching pathway. We defined nine Branch categories: cytonuclear proteostasis, ER proteostasis, mitochondrial proteostasis, nuclear proteostasis, translation, proteostasis network (PN) regulation, extracellular proteostasis, the ALP, and the UPS (Figure 1). Note that the term "cytonuclear" refers to components that support proteostasis in both the cytosol and the nucleus, whereas "nuclear" designates components dedicated primarily to the nucleus (e.g.,

histone chaperones). "PN regulation" refers to components that either control transcription of proteostasis network component genes (e.g., heat shock factor 1, or HSF1) or control translation as part of, for example, a cell stress response (e.g., PERK and the other kinases that activate the integrated stress response). The next level of the taxonomy scheme is Class, which refers to a component's function in proteostasis (e.g., chaperones, protein transport, etc.). The Classes associated with each Branch are shown in Figure 2. The remaining annotation levels, Group, Type, and Subtype, provide increasingly specific descriptors of proteostasis functions within a Class. Our goal was to be parsimonious in our annotations and use only as many descriptors as were necessary to give a basic understanding of a component's role in proteostasis. Thus, not every component has Type or Subtype annotations. Some components have multiple roles in the proteostasis network. These are given multiple entries in our list to reflect each role. For example, the many roles of VCP in proteostasis, many of which are noted above, have resulted in its having eight annotations.

As an illustrative example, we applied our taxonomic scheme to BAG family molecular chaperone regulator 1, or BAG1, a cytonuclear BAG-domain-containing nucleotide exchange factor for HSP70. As shown in Figure 3, this yielded the following annotations: Branch: cytonuclear proteostasis; Class: chaperones; Group: HSP70 system; Type: HSP70 nucleotide exchange factor; Subtype: BAG domain family. Note that annotations are in general not unique to individual proteostasis network components. For example, the four remaining BAG-domain-containing nucleotide exchange factors in humans (BAG2-BAG5) all have the same set of annotations as BAG1 (Figure 3). BAG6, which has in the past been included as an HSP70 nucleotide exchange factor, actually does not have a canonical BAG domain, is not active as a nucleotide exchange factor, and so was not given this set of annotations (although it does have other roles in proteostasis) [35].

Distribution of components in the human proteostasis network

The complete list of proteostasis network components is presented in Table S1, which includes the ALP and the UPS, although these latter two systems are described separately (see the Data Availability Statement at the end of the main text for a link to download Table S1). There are 4,002 entries in the proteostasis network representing 3,125 unique components corresponding to unique genes. These entries are divided among the Branches as shown in Figures 1 and 2. The

ALP consists of 1,048 entries representing 856 unique components, while the UPS consists of 1,693 entries representing 1,411 unique components. These two Branches are by far the largest in the proteostasis network, and they have substantial overlap with each other, with 156 of their components being listed in both Branches. The remaining seven Branches, which are the focus in this manuscript, together consist of 1,261 entries representing 1,192 unique components, of which 67 are also in the ALP (but not the UPS), 85 are also in the UPS (but not the ALP), and 26 are also in both the ALP and the UPS.

All 1,261 entries in the non-ALP/UPS proteostasis network have Class annotations, divided among 15 unique categories. Four of these Class categories are used in multiple Branches. For example, the Class annotation "chaperone" applies in the cytonuclear, nuclear, ER, mitochondrial, and extracellular Branches. This multiplicity reflects the presence of distinct complements of chaperones that have analogous functions in the compartments (subcellular or extracellular) in which they reside. The Class annotations "folding enzyme", "protein transport", and "organelle-specific protein degradation" also appear in multiple Branches and likewise represent sets of components with corresponding proteostasis functions in different compartments. The distribution of Classes among the Branches is illustrated in Figure 2.

All 1,261 entries also have Group annotations, divided among 73 categories. Most entries (994 out of 1,261) have Type annotations, divided among 110 categories, but Subtype annotations were only necessary for a minority of entries (457 out of 1,262), divided among 81 categories.

The seven Branches covered in this manuscript can be thematically divided into the compartment-based Branches—the cytonuclear, ER, mitochondrial, nuclear, and extracellular proteostasis Branches—the translation Branch, and the PN regulation Branch. The annotation of each is described below.

Annotation of the compartment-based Branches of the proteostasis network

To demonstrate how Group, Type, and Subtype annotations were assigned in the compartment-based proteostasis network Branches, we show how the cytonuclear chaperones are distributed within these finer-grained categories in Table 1. The cytonuclear chaperones (Branch: cytonuclear proteostasis, Class: chaperones) are divided into six Groups. Four of these Groups consist of the components of structurally and mechanistically distinct chaperone systems that have broad substrate specificity: the HSP70 system, the CCT/TRiC system, the sHSP (small heat shock

protein) system, and the HSP90 system. In addition, one Group serves as a bridge between two of the preceding systems, the HSP70-HSP90 system integration Group. The final Group is "chaperone for alpha-actinin in myofibril assembly", which contains a chaperone (NRAP) for a specific and important substrate in muscle cells (alpha-actinin). Each of these Groups is then divided into the Types of components that exist within the Group. Thus, for example, there are four Types in the HSP70 system Group [28]: the HSP70 chaperones, the J-domain containing cochaperones, the HSP70 nucleotide exchange factors, and the HSP70 nucleotide exchange inhibitors. Similarly, the CCT/TRiC, sHSP, HSP90 system, HSP70-HSP90 system integration, and chaperone for alpha-actinin in myofibril assembly Groups contain three, two, two, one, and zero Types respectively (Table 1). In the chaperone system Groups (HSP70, CCT/TRiC, sHSP, and HSP90), one of the Types contains the Group's parent chaperone (HSP70s, CCT/TRiC subunits, sHSPs, or HSP90s). The other Types contain cochaperones, other accessory proteins, or in the CCT/TRiC system Group, components that have sequence similarity to the parent chaperone but are not part of the CCT/TRiC complex. This last Type ("sequence similar non-CCT/TRiC subunit") was included in the proteostasis network via the domain-based criterion, but also because there is evidence that the three members of this Type form a complex with CCT/TRiC components that mediates assembly of the BBSome complex [36]. In the HSP70-HSP90 system integration Group, the single Type indicates that the members of this Group are all cochaperones for both HSP70 and HSP90. Finally, the Chaperone for alpha-actinin in myofibril assembly Group has only one component and therefore does not need Type or Subtype annotations.

Some of the Types in Table 1 (five out of twelve) can be further refined with Subtype annotations based on structural or functional criteria or both. The nucleotide exchange factor Type of the HSP70 system Group consists of three clear Subtypes based on their structures [37]: the HSP110/GRP170 subtype, whose members consist of HSP70-like nucleotide- and substrate-binding domains fused to an α-helix bundle domain via an insertion of variable length (3 members); the BAG domain subtype, whose members contain a BAG domain (5 members); and the HSPBP1 subtype, whose members contain armadillo repeats domain (1 member). There are four Subtypes of CCT/TRiC cochaperones, two of which are characterized by their functions—actin and G-protein folding (5 members) [38] and tubulin folding (5 members) [39]—one of which consists of subunits of the prefoldin complex (9 members) [40,41], and one of which is a newly identified cochaperone that binds to the open state of CCT/TRiC (1 member) [42]. There are seven

Subtypes of HSP90 cochaperones whose members are characterized by the presence, absence, or combinations of CS, TPR/CC-TPR, PPIase, and/or RPAP3-like domains [43-45]. Finally, there are 12 Subtypes of the HSP70-HSP90 joint cochaperones defined by the domains that they contain in addition to the CC-TPR domain [31].

Assignment of annotations in the other compartment-based Branches were done similarly. However, some Branches have proteostasis mechanisms that could not be divided by the structures of the components like the chaperones can be. To illustrate how the annotation of such components was handled we focus on the protein transport Class annotations of the ER, mitochondrial, and nuclear Branches. The relevant Group, Type, and Subtype annotations for the protein transport Class in these Branches are shown in Table 2. These components were generally categorized by their membership in key complexes, by their participation in a particular pathway, or by their function. Thus, in the protein transport Class of the ER proteostasis Branch, there are complex-based Groups like SEC61 channel complex component [46], signal recognition particle component [47], TRAP complex component [48], among others; pathway-based Groups like GET pathway component [49] or ER-Golgi trafficking [50-56]; and function-based Groups like ER signal peptidase. In the nuclear proteostasis Branch, the complex-based Group was nuclear pore complex [57,58]; the pathway-based Group was nuclear transport, RAN system; and the function-based Group was nuclear transport receptor [59,60], which was further divided into Types by whether they were importins of beta-type, alpha-type, or neither type. For the mitochondrial proteostasis Branch, the components in the protein transport Class were divided into Groups according to whether they are involved in import across the outer or the inner mitochondrial membrane and then divided into Types according to their membership in complexes, like the TOMM complex, or the TIMM17/23 complex [61-64].

Annotation of the translation Branch of the proteostasis network

The translation Branch of the proteostasis network must encompass all the protein synthesis that occurs in an organism. Thus, it is divided into two Classes, cytosolic translation and mitochondrial translation. Each of these Classes is divided into Groups for their respective ribosomes (the cytoribosome and the mitoribosome) [65,66] and their rather extensive sets of biogenesis factors [67-70], which are required for them to mature and assemble properly. The RNA-based components of the ribosome are not included, however. Each Class also has Groups

for the steps of translation: translation initiation [71,72], elongation [72-75], and termination [72,76]. The translation elongation Group includes the tRNA synthetases (which overlap to a considerable extent between the cytosolic and mitochondrial translation systems) because the charged tRNAs are the proximal substrates for protein synthesis [73,75]. It does not, however, include the pathways for the synthesis of the amino acids that are loaded onto the tRNAs, as we view these as essentially metabolic processes. In addition to these Groups, the cytosolic translation Class has two additional Groups. The first is termed nascent peptide husbandry, and it includes components that chaperone nascent proteins (the RAC [77] and NAC [78] complexes), those that cleave N-terminal methionine residues, and those that acetylate or myristoylate the N-terminus [78]. The second Group is for the ribosome-associated quality control system, and it contains the components involved in ribosomal rescue and disposal of unfinished peptides from stalled ribosomes [79-81].

Annotation of the PN regulation Branch of the proteostasis network

The expression levels of the components of some proteostasis network Branches are in some cases regulated by prominent stress-responsive transcription factors, like the heat shock factors (primarily HSF1) [82] for the cytosolic proteostasis Branch, or XBP1 and ATF6 for the ER proteostasis Branch [83]. However, except for the ALP Branch, we chose not to include these regulatory elements in their corresponding Branches because in many cases their activities transcend individual Branches. For example, the mitochondrial HSP60/10 chaperones (HSPD1 and HSPE1) and the extracellular chaperone clusterin (CLU) are both regulated by HSF1 [84,85]. Thus, we created a separate Branch for proteostasis regulators. This includes transcription factors as well as components that control transcription factor activity, like IRE1/ERN1 for XBP1 [83] and KEAP1 for NRF2/NFE2L2 [86]. It also includes components that can control translation in response to a variety of stresses including some microRNAs [87]. However, the most prominent examples of translational control are the kinases that activate the integrated stress response: EIF2AK1/HRI, EIF2AK2/PKR, EIF2AK3/PERK, and EIF2AK4/GCN2 [88].

The regulators of the ALP were listed in the ALP Branch rather than the proteostasis regulators Branch because we thought that it was important for the ALP to be self-contained. By including them in a Class within the ALP Branch, the regulators of autophagy gene expression

maintain a close and prominent association with the ALP. This is particularly important for the transcription factors like ATF4, which regulate both the ALP and other aspects of proteostasis.

Discussion

We have defined the boundaries of the human proteostasis network and presented a curated and manually annotated list of the cellular components responsible for maintaining proteostasis in compartments, represented in the cytonuclear, ER, mitochondria, nuclear, and extracellular Branches of the proteostasis network; for protein biogenesis, represented in the translation Branch; and for regulating the other Branches, represented in the PN regulation Branch. This corresponds to 1,261 genes, or about 5% of human protein-coding genes. This level of investment by evolution into proteostasis illustrates the intricacies of the problem of maintaining proteostasis, which becomes more acute as organisms become more complex. This observation becomes even clearer in the next installments in this series, in which we will enumerate the components of the ALP and the UPS. Each of these degradative pathways are approximately the same size as the subset of proteostasis network genes described herein, illustrating just how finely tuned are the processes for protein turnover.

By providing a comprehensive annotation of the human proteostasis network, we aim to make this profoundly important biological system accessible for routine incorporation into studies across diverse biological and disease contexts. With such a comprehensive list, it will now be possible to assess, for example, which components of the proteostasis network that are important for robustness and stress resilience in youth lose capacity in aging. Likewise, assessments of the proteostasis network across the range of human diseases will enable novel insights to be made on protective mechanisms, basis of failure, and adaptive mechanisms. This, in turn, will better identify potential targets for detection and therapeutic approaches.

Furthermore, we aim to facilitate the identification of systems and processes that were not known previously to engage the proteostasis network by enabling others to view their experimental systems through the lens of proteostasis. For example, the overlaps between a list of hits from a biochemical or bioinformatic screen and the proteostasis network could inform researchers from other fields on the contribution (or lack thereof) of proteostasis network members to the process that is the subject of the screen. Finally, we consider this to be an ongoing effort, and we welcome

community feedback on the composition of the proteostasis network, which is also shared via the Proteostasis Consortium website (https://proteostasisconsortium.com) and through an interactive web resource (https://proteostasisnetwork.streamlit.app/). Please contact us at pnannotation@gmail.com with suggestions for components that we have excluded that should have been included, or vice versa. We regard this list as the first version of the proteostasis network, which will evolve as new experimental data on human proteostasis becomes available.

Methods

Criteria for inclusion in the proteostasis network

We used two criteria to decide whether a given macromolecule should be in the proteostasis network. The first was the "entity-based" criterion: an "entity" (an individual protein or non-coding RNA, or a complex) was included as a member of the proteostasis network simply if there was specific evidence in the literature that it has a role in proteostasis. The second was the "domain-based" criterion: a protein was included as a member of the proteostasis network even if there was no literature evidence that it had a role in proteostasis if it contained a domain or domains that were otherwise very strongly associated with proteostasis. This latter criterion was important for the inclusion of uncharacterized proteins that are clearly members of structural families that usually have roles in proteostasis. See the Results section for examples of how these criteria were applied.

Table 3 shows the literature sources that we primarily used to build our preliminary lists of proteostasis network components for the various systems mentioned above. We also used annotation databases such as the Gene Ontology Resource [34], the Reactome Pathway Database,[89] and the Kyoto Encyclopedia of Genes and Genomes (KEGG) [90], as well as more general databases like UniProt [91] and InterPro [92] to supplement the information from the references in Table 3. The preliminary lists generated from these sources were subsequently vetted gene by gene by the authors. Members of the Proteostasis Consortium with relevant subject area expertise presented rationales for inclusion or exclusion of borderline genes, and final decisions on inclusion or exclusion were made by group consensus. The full annotation is presented in Table S1 (see the Data Availability Statement at the end of the main text for a link to download Table S1).

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Disclosure statement

The authors declare no competing interests.

Data availability statement

Supplementary Information can be found on the Proteostasis Consortium website (privately-advertised link): https://sites.northwestern.edu/proteostasis-institute/data-download/. Navigate to the header "Manuscript 1 / Survey of the Human Proteostasis Network: Translation, Protein Folding, and Organelle-Specific Systems." This study references Tables S1. The related survey of the ALP also references a single table (Table S1), while the survey of the UPS references seven (Tables S1-S7). The annotation in Table S1 is precisely the same for each study, varying only in which Branches are additionally displayed for the convenience of the reader. Prior to publication, this information will be supplied through the Dryad data repository.

Supplementary Information

Table S1. Human Proteostasis Network Annotation

The complete annotation of the Human Proteostasis Network is presented in the Main tab. The annotation of the Branches covered in this manuscript begins at Row 3 in the Main tab and goes through Row 1501. Each Branch is additionally presented by itself in the <Branch>_only tabs. Detailed information about the tabs and fields is given in the README tab.

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Tables

Table 1. Groups, Types, and Subtypes within the cytonuclear chaperone Class.

Group	Type	Subtype	Components
HSP70[a] system	HSP70	(no Subtype)	9
	J-domain containing HSP70 cochaperone	(no Subtype)	26
	HSP70 nucleotide exchange factor	HSP110/GRP170[e] subtype	3
		BAG[f] domain subtype	5
		HSPBP1[g] subtype	1
	HSP70 nucleotide exchange inhibitor	TPR[h] domain subtype	1
CCT/TRiC[b] system	CCT/TRiC subunit	(no Subtype)	10
	Sequence similar non-CCT/TRiC subunit	(no Subtype)	3
	CCT/TRiC cochaperone	Phosducin-like proteins	5
		Tubulin folding	5
		Open state cofactor	1
		Prefoldin subunit	9
sHSP[c] system	Small HSP	(no Subtype)	10
HSP90[d] system	HSP90	(no Subtype)	2
	HSP90 cochaperone	CS[i] domain containing	16
		TPR domain containing	5
		TPR and CS domain containing	1
		CC-TPR[j] and Armadillo domain containing	2
		TPR and PPIase[k] domain containing	4
		CC-TPR and RPAP3[l]-like C-terminal domain containing	1
		No characteristic domain	9
HSP70-HSP90 system integration	HSP70-HSP90 joint cochaperone	CC-TPR and Armadillo domain containing	1
		CC-TPR and Cns1/TTC4[m] wheel domain containing	1
		CC-TPR and CS domain containing	1
		CC-TPR and J domain containing	1
		CC-TPR and methyltransferase domain containing	1
		CC-TPR and phosphatase domain containing	1
		CC-TPR and PPIase domain containing	4
		CC-TPR and RPAP3-like C-terminal domain containing	1
		CC-TPR and SGTA[n] dimerization domain containing	2
		CC-TPR and STI/HOP[o] domain containing	1
		CC-TPR domain and Ub ligase[p]	4
		CC-TPR domain containing	6

Chaperone for alpha-actinin in myofibril assembly	(no Type)	(no Subtype)	1

[a]HSP70: heat shock protein 70. [b]CCT: chaperonin containing TCP1, TRiC: T-complex protein ring complex. [c]sHSP: small heat shock protein. [d]HSP90: heat shock protein 90. [e]HSP110: heat shock protein 110, GRP170: glucose regulated protein 170. [f]BAG: BCL2-associated athanogene. [g]HSPBP1: HSPA binding protein 1. [h]TPR: tetratricopeptide repeat domain; [i]CS: CHORD and SGT1 domain; [j]CC-TPR: carboxylate clamp TPR; [k]PPIase: peptidylprolyl isomerase; [l]RPAP3: RNA polymerase II-associated protein 3; [m]Cns1/TTC4: tetratricopeptide repeat/cyclophilin seven suppressor 1; [n]SGTA: small glutamine-rich tetratricopeptide repeat-containing protein alpha; [o]STI/HOP: stress-induced phosphoprotein/HSC70HSP90-organizing protein; [p]Ub ligase: ubiquitin ligase.

Table 2. Groups, Types, and Subtypes within the protein transport Classes of the ER, mitochondrial, and nuclear proteostasis Branches.

Group	Type	Subtype	Components
Branch: ER proteostasis; Class: Protein transport			
SEC61 channel complex component	(no Type)	(no Subtype)	6
Signal recognition particle component	(no Type)	(no Subtype)	6
SRP[a] receptor subunit	(no Type)	(no Subtype)	2
TRAP[b] complex component	(no Type)	(no Subtype)	4
SEC61 channel accessory protein	(no Type)	(no Subtype)	5
ER signal peptidase	(no Type)	(no Subtype)	5
GET[c] pathway component	(no Type)	(no Subtype)	8
Transmembrane protein import	GEL[d] complex component	(no Subtype)	2
	BOS[e] complex component	(no Subtype)	5
	PAT[f] complex component	(no Subtype)	2
	EMC[g] complex component	(no Subtype)	10
Removal of misinserted transmembrane proteins	(no Type)	(no Subtype)	1
ER-Golgi trafficking	Anterograde transport	COPII[h] vesicle component	12
	Anterograde transport	SAR1A/B[i] guanine nucleotide exchange factor	1
	Retrograde transport	COPI[j] vesicle component	9
	Retrograde transport	COPI coating and uncoating	2
	Retrograde transport	COPI scaffolding in the Golgi	3
	Retrograde transport	Receptor for KDEL[k]-containing proteins for retrograde transport	4
	Antero- and retrograde transport	Cargo receptor	9
Branch: Mitochondrial proteostasis; Class: Protein transport			
Outer membrane import	TOMM[l] complex	(no Subtype)	9
Inner membrane import	TIMM17/23[m] complex	(no Subtype)	8
	TIMM22[m] complex	(no Subtype)	3
	For assembly of inner membrane complexes	(no Subtype)	1
Cytosolic mitochondria import stimulation factor	(no Type)	(no Subtype)	2
Branch: Nuclear proteostasis; Class: Protein transport			
Nuclear pore complex	(no Type)	(no Subtype)	36
Nuclear transport receptor	Importin, beta type	(no Subtype)	20
	Importin, alpha type	(no Subtype)	7
	Non-importin transport receptor	(no Subtype)	3
Nuclear transport, RAN[n] system	(no Type)	(no Subtype)	14

[a]SRP: signal recognition particle; [b]TRAP: translocon-associated protein; [c]GET: guided entry of tail-anchored proteins; [d]GEL: GET- and EMC-like; [e]BOS: Back of SEC61; [f]PAT: protein associated with translocon; [g]EMC: ER membrane protein complex; [h]COPII: coat protein II; [i]SAR: secretion-associated Ras-related GTPase; [j]COPI: coat protein I; [k]KDEL: amino acid sequence Lys-Asp-Glu-Leu; [l]TOMM: translocase of the outer membrane of mitochondria; [m]TIMM: translocase of the inner membrane of mitochondria; [n]RAN: Ras-related nuclear protein.

Table 3. Primary literature sources for proteostasis network components.

	Subject	References
Protein synthesis	Ribosome	[65,66]
	Ribosome biogenesis	[67-70]
	Translation initiation	[71,72]
	Translation elongation	[72-75]
	Translation termination	[72,76]
	Nascent peptide husbandry	[77,78]
	Ribosomal quality control	[79-81]
Chaperones and folding enzymes	General	[14-16]
	HSP70 system	[28,30,37,93,94]
	HSP90 system	[43-45]
	TRiC/CCT system	[38-40,95,96]
	sHSP	[97-99]
	PDI	[100]
	PPIase	[101]
Endoplasmic reticulum	Import	[46-49,102-104]
	Export	[50-56]
	Glycosylation/glycoprotein folding	[105-107]
	ERAD	[107-111]
	Collagen	[112]
Mitochondria	Import	[61-64,113,114]
	Degradation	[24,115-117]
Nucleus	Histone chaperones	[118,119]
	Nuclear pore complex	[57,58]
	Nuclear transport	[59,60]
PN regulation	Transcription factors	[20,82,86,88,120-123]
	Other regulators	[87]

Figure Legends

Figure 1. The proteostasis network. Cartoon representation of a eukaryotic cell with all nine Branches of the proteostasis network in our taxonomic system, shown in boxed labels and positioned according to their primary subcellular localization. The number of components and entries is indicated for the Branches covered in this manuscript. Components correspond to unique genes, whereas entries represent annotation instances; individual components may be assigned to multiple proteostasis network categories and therefore contribute to more than one entry. Created in part in BioRender.

Figure 2. Branches and Classes of the human proteostasis network. The nine Branches of the proteostasis network with their cognate Classes. Each Class partitions into Groups, and in some cases, those are elaborated as Types and Subtypes. The complete hierarchy is available in Table S1.

Figure 3. The hierarchical annotation framework. Classification of Branch, Class, Group, Type, and Subtype illustrated using the HSP70 nucleotide exchange factor BAG1 as an example.

Figures.

Figure 1. The proteostasis network.

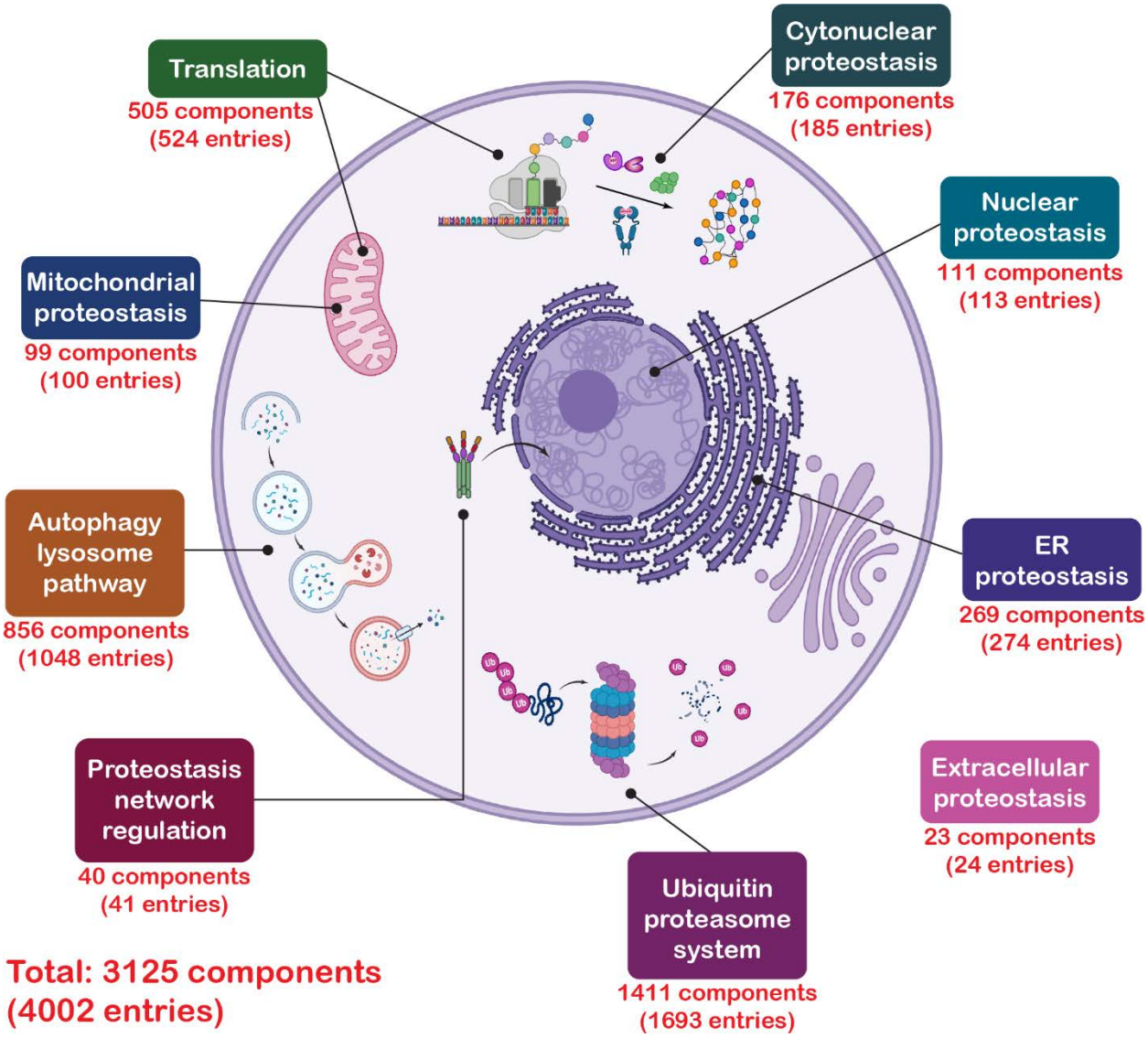

Figure 2. Branches and Classes of the human proteostasis network.

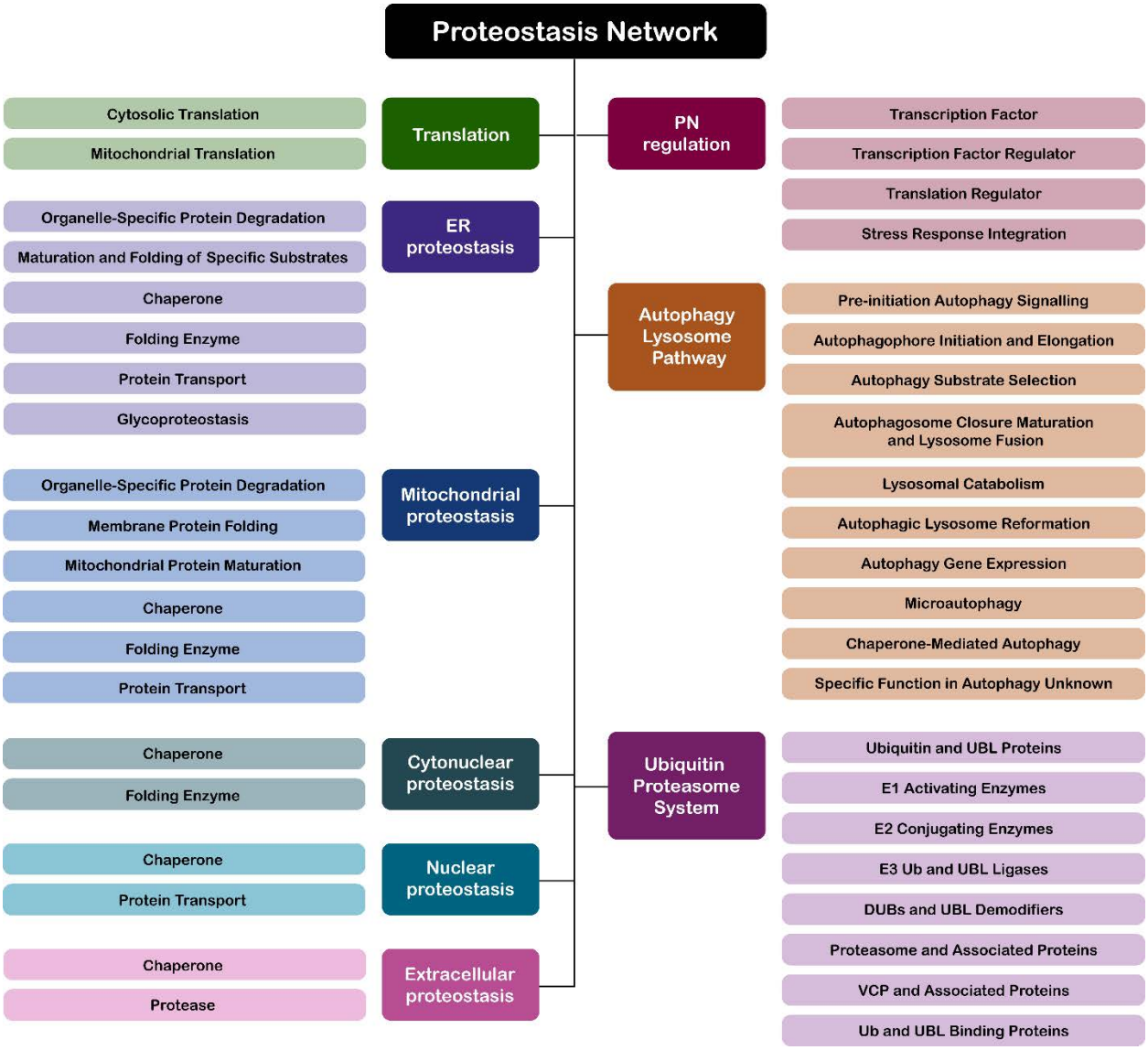
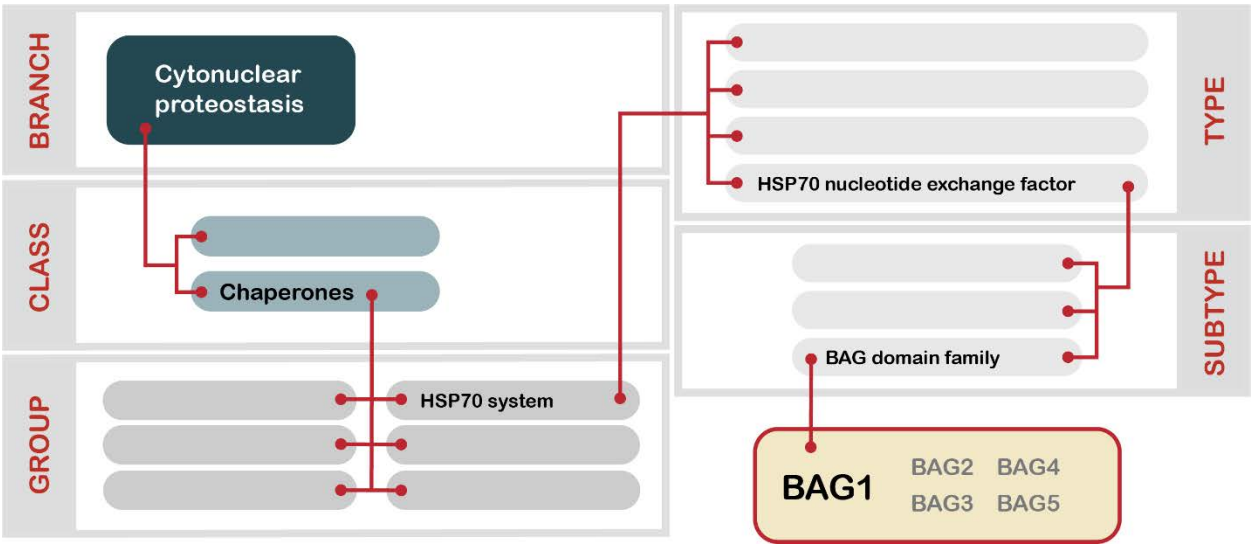
Figure 3. The hierarchical annotation framework.